

Numerical Analysis PhD Exam, May 2001

Do any 8 of the following 10 problems.

Part 1: Numerical Linear Algebra

1. This problem has five parts.

- (a) Under what conditions does a real n by n matrix A have a Schur decomposition $A = U^T T U$, where U and T are real n by n matrices with U orthogonal and T upper triangular?
- (b) Under what conditions on A is T diagonal?
- (c) Does the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

- have an orthogonal set of eigenvectors?
- (d) Give a careful statement of the singular value decomposition of a real m by n matrix.
- (e) Under what conditions does the singular value decomposition exist?

2. Consider the linear system $Ax = b$.

- (a) If the rows of A are linearly independent, derive a formula for the solution of minimal 2-norm.
- (b) If the columns of A are linearly independent, derive a formula for the x that minimizes the 2-norm of the residual $r = b - Ax$.
- (c) Use the singular value decomposition of A to give a formula for the x that minimizes the 2-norm of the residual, and among all the x 's which minimize the 2-norm, it has minimal norm.

3. This problem has two parts.

- (a) Given a vector $x \in \mathbb{R}^n$ and a natural number $k < n$, give a formula for a unit vector w for which the vector $y = (I - 2ww^T)x$ satisfies the following conditions: $y_i = x_i$ for $i < k$ and $y_i = 0$ for $i > k$.
- (b) Using these Householder transformations, write a pseudo code (or a matlab code) for reducing a real symmetric matrix to tridiagonal form using Householder similarity transformations.

4. This problem has three parts.

- (a) State and prove the Gershgorin Circle Theorem.
- (b) Let

$$A = \begin{pmatrix} 5 & 1 & 3 \\ 2 & 4 & 1 \\ 3 & -1 & 4 \end{pmatrix}.$$

What can you say about the location of the eigenvalues of A ?

- (c) Use the Gershgorin Circle Theorem to estimate the size of the largest root of the polynomial $p(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0$.

5. This problem has two parts.

- (a) Find the ellipse or hyperbola, of the form $ax^2 + by^2 = 1$, that best fits n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ in the plane by computing the least squares solution to the n linear equations gotten by substituting each of the data points into the quadratic equation.
- (b) Test your method with the dataset $(1, 1), (2, 4), (3, 9), (4, 16)$.

Part II: Numerical Analysis

6. This problem has three parts.

- (a) State Newton's method (the algorithm) for solving $f(x) = 0$ where $f : \mathbb{R} \rightarrow \mathbb{R}$.
- (b) Assuming f is smooth and we have an initial guess sufficiently close to a root p , state sufficient condition(s) for the method to converge quadratically.
- (c) Show that Newton's method applied to $x^5 = 2$ converges quadratically to the root $2^{1/5}$ from any starting guess $x > 0$.

7. This problem has three parts.

- (a) Given a function f with $n + 1$ continuous derivatives on the interval $I = [-1, 1]$, show that, given $n + 1$ points $\{x_0, x_1, \dots, x_n\}$ in I , there exists a unique polynomial $P_n(x)$ of degree $\leq n$ such that

$$P_n(x_i) = f(x_i) \quad \text{for } i = 0, \dots, n.$$

- (b) Prove that, given $t \in I$, there exists $\eta \in I$ such that

$$f(t) - P_n(t) = \frac{(t - x_0) \cdots (t - x_n)}{(n + 1)!} f^{(n+1)}(\eta).$$

- (c) From the above we get that (all norms are on the interval I)

$$\|f - P_n\|_\infty \leq \max_{t \in I} |(t - x_0)(t - x_1) \cdots (t - x_n)| \frac{\|f^{(n+1)}\|_\infty}{(n + 1)!}.$$

What choice for the points $x_0, \dots, x_n$ minimizes the right hand side of this error bound?

8. This problem has two parts.

- (a) Find a function $q(x)$ which has a continuous first derivative for each $x \in (-\infty, \infty)$ with the properties:

$$q(x) = \begin{cases} 1, & \text{if } x = 0, \\ 0, & \text{for all } |x| \geq 2, \\ a_0 + a_1x + a_2x^2, & \text{for all } x \in [-2, -1], \\ b_0 + b_1x + b_2x^2, & \text{for all } x \in [-1, 1], \\ c_0 + c_1x + c_2x^2, & \text{for all } x \in [1, 2]. \end{cases}$$

- (b) Use the function $q(x)$ defined in part a to interpolate a set of equally spaced points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, where $h = x_{k+1} - x_k$ for all k . In particular, describe how to find constants c_k so that the function

$$Q(x) = \sum_{k=0}^n c_k q\left(\frac{x - x_k}{h}\right)$$

has the property that $Q(x_k) = y_k$ for all k .

9. Let $P_2(x)$ be a quadratic polynomial interpolating $g(x)$ at $x = 0, h, 2h$.

(a) Use this to derive a numerical integration formula I_h for

$$I = \int_0^{3h} g(x) dx.$$

(b) Use a Taylor series expansion of $g(x)$ to show

$$I - I_h = \frac{3}{8}h^4 g^{(3)}(0) + O(h^5).$$

10. Triple Recursion Formula. If $\{\phi_n(x)\}$ is an orthogonal family of polynomials on $[a, b]$, with respect to the weight function $w(x) \geq 0$, and $n \geq 1$, then show that

$$\phi_{n+1}(x) = (a_n x + b_n)\phi_n(x) - c_n \phi_{n-1}(x),$$

for some constants a_n, b_n , and c_n . (Hint: Let $g(x) = \phi_{n+1}(x) - a_n x \phi_n(x)$, where a_n is a constant chosen so that $g(x)$ has degree n .)